

SQL Data Analysis Report: E-Commerce Sales Dataset

Executive Summary

This project analyzes an e-commerce sales dataset containing 1,200 transactions recorded between January 2023 and June 2025. Using SQL, the dataset was explored, validated, and analyzed to identify trends in sales performance, customer behavior, product demand, payment preferences, and marketing effectiveness.

The analysis revealed total sales revenue of \$1,264,761.96, with revenue distributed across multiple product categories. Chairs emerged as the highest revenue-generating product category, while Instagram was identified as the strongest customer acquisition channel. Additionally, a small group of customers contributed significantly to overall revenue, highlighting opportunities for customer retention initiatives.

The project demonstrates practical SQL skills including data validation, filtering, aggregation, grouping, ranking functions, subqueries, and business intelligence reporting.

1. Dataset Overview

The dataset contains transactional records from an e-commerce business and includes information related to customers, products, payments, referrals, and order fulfillment.

Dataset Characteristics

- Total Records: 1,200
- Analysis Period: January 1, 2023 to June 30, 2025
- Analysis Tool: MySQL

Key Fields

- OrderID
- CustomerID
- Product
- Quantity
- UnitPrice
- TotalPrice
- PaymentMethod
- OrderStatus
- ReferralSource
- Date

2. Data Quality Assessment

Before conducting business analysis, the dataset was evaluated for accuracy and completeness.

Findings

- Total records analyzed: 1,200
- No duplicate OrderIDs detected
- No missing values identified in critical fields
- Date records were consistent throughout the analysis period
- Revenue fields contained valid numerical values

Business Impact

High-quality data is essential for reliable business intelligence. The absence of duplicates and missing values increases confidence in the accuracy of the analytical findings.

3. Revenue Performance Analysis

Total Revenue

The company generated total sales revenue of:

\$1,264,761.96

during the reporting period.

Analysis

This revenue level reflects substantial customer purchasing activity throughout the analysis period and provides a strong basis for evaluating product and customer performance.

Business Insight

Revenue is distributed across several product categories, reducing dependence on a single source of income and improving business stability.

4. Product Performance Analysis

Objective

Identify products generating the highest revenue.

Results

Product	Revenue
Chair	\$195,620.11
Printer	\$195,612.61
Laptop	\$192,126.56

Analysis

Chairs generated the highest revenue, narrowly outperforming Printers and Laptops. Revenue distribution was relatively balanced, indicating diversified customer demand.

Business Recommendation

Management should maintain adequate inventory levels for top-performing products while exploring promotional opportunities for lower-performing categories.

5. Order Status Analysis

Objective

Evaluate revenue distribution across order fulfillment statuses.

Results

Status	Revenue
Cancelled	\$276,396.21
Pending	\$256,328.15
Other Statuses Remaining Revenue	

Analysis

Cancelled orders contributed the highest order value in the dataset. Since cancelled transactions typically do not represent realized revenue, this suggests that the dataset records order values at the point of purchase rather than after successful fulfillment.

Business Recommendation

Future financial reporting should distinguish between gross order value and realized revenue by excluding cancelled transactions from revenue calculations.

6. Payment Method Analysis

Objective

Understand customer payment preferences.

Results

Payment Method	Revenue
Credit Card	\$263,847.63
Online Payment	\$262,442.94
Other Methods	Remaining Revenue

Analysis

Revenue distribution across payment methods is relatively balanced, suggesting customers are comfortable using multiple payment options.

Business Recommendation

Maintaining a variety of payment methods can improve customer convenience and reduce checkout abandonment.

7. Monthly Revenue Trend Analysis

Objective

Identify revenue patterns over time.

Key Findings

- Highest Revenue Month: June 2024
- Revenue Generated: \$68,068.54
- Strong Performance Month: May 2023
- Revenue Generated: \$63,836.84

Analysis

Revenue fluctuated throughout the reporting period, suggesting the influence of seasonal demand, promotional campaigns, or changing customer purchasing behavior.

Business Recommendation

Further investigation into seasonal patterns can improve forecasting and inventory planning.

8. Customer Value Analysis

Objective

Identify high-value customers.

Top Customers

Customer ID Revenue

C38840 \$5,723.23

C57276 \$3,456.40

C67260 \$3,390.80

Analysis

A relatively small group of customers generated significantly higher revenue than the average customer, indicating opportunities for targeted retention strategies.

Business Recommendation

Implement loyalty programs, personalized offers, and exclusive promotions for high-value customers to maximize customer lifetime value.

9. Referral Source Performance Analysis

Objective

Evaluate the effectiveness of customer acquisition channels.

Results

Referral Source	Revenue
Instagram	\$275,285.45
Email	\$261,808.55
Google	\$250,441.48
Facebook	\$250,410.90
Referral	\$226,815.58

Analysis

Instagram generated the highest revenue contribution among all referral channels, while Email marketing also delivered strong performance.

Business Recommendation

Marketing investments should continue supporting Instagram and Email campaigns while monitoring conversion rates and acquisition costs to determine long-term profitability.

10. Customer Revenue Ranking

Objective

Rank customers based on total spending using SQL Window Functions.

Top 5 Customers

	Rank	Customer ID	Revenue
1	C38840	\$5,723.23	
2	C57276	\$3,456.40	
3	C67260	\$3,390.80	
4	C13877	\$3,384.90	
5	C18404	\$3,370.20	

Learning Outcome

This analysis demonstrates the use of SQL RANK() window functions for customer segmentation and performance analysis.

11. Product Revenue by Payment Method

Objective

Analyze purchasing behavior across payment methods.

Analysis

Different payment methods exhibit varying purchasing preferences across product categories.

Business Insight

Customer purchasing behavior differs based on payment method, providing opportunities for targeted promotions and personalized marketing campaigns.

Business Recommendation

Marketing teams can tailor promotions according to customer payment preferences to improve engagement and sales performance.

12. Average Order Value (AOV) Analysis

Objective

Determine the average amount spent per transaction.

Analysis

Average Order Value (AOV) measures the typical amount customers spend during a single purchase.

Business Insight

AOV is a critical e-commerce metric used to evaluate customer purchasing behavior and overall sales performance.

Business Recommendation

Cross-selling, upselling, and product bundles can help increase average order value and overall revenue.

13. Order Status Distribution Analysis

Objective

Evaluate the number of orders within each order status category.

Analysis

This analysis provides insight into operational performance by showing how orders are distributed across Delivered, Shipped, Pending, and Cancelled statuses.

Business Insight

A high volume of cancelled or pending orders may indicate operational challenges that require management attention.

Business Recommendation

Monitoring order status distribution can help identify fulfillment bottlenecks and improve customer satisfaction.

14. Top-Selling Products by Quantity

Objective

Identify the products purchased most frequently by customers.

Analysis

Sales volume reveals product popularity, while revenue highlights profitability.

Business Insight

Products with high sales volume may not necessarily generate the highest revenue, making both metrics important for decision-making.

Business Recommendation

High-volume products should be prioritized for inventory management and stock planning.

15. Product Revenue Contribution Analysis

Objective

Determine the percentage contribution of each product category to total company revenue.

Analysis

This analysis evaluates how much each product contributes to overall business revenue and identifies the organization's primary revenue drivers.

Business Insight

Products contributing a larger share of total revenue should receive strategic attention in pricing, inventory management, and promotional planning.

Business Recommendation

Management should focus investment and marketing efforts on high-contributing product categories while developing growth strategies for lower-performing products.

16. SQL Techniques Applied

The following SQL concepts were utilized throughout the project:

- SELECT Statements
 - WHERE Clauses
 - ORDER BY
 - GROUP BY
 - COUNT()
 - SUM()
 - AVG()
 - Aggregate Functions
 - Subqueries
 - Window Functions (RANK)
 - Business Intelligence Reporting
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Conclusion

This project successfully applied SQL techniques to transform raw transactional data into meaningful business insights.

Key findings include:

- Total Revenue: \$1,264,761.96
- Highest Revenue Product: Chair (\$195,620.11)
- Top Customer: C38840 (\$5,723.23)
- Strongest Referral Channel: Instagram (\$275,285.45)
- Highest Revenue Month: June 2024 (\$68,068.54)
- Leading Payment Method: Credit Card (\$263,847.63)

The analysis demonstrates how SQL can support data-driven decision-making by identifying revenue drivers, customer trends, product performance, and marketing effectiveness. These insights can help management improve customer retention, optimize marketing investments, strengthen operational efficiency, and support future business growth.